

LM3880 Simple Triple Rail Power Supply Sequencer

1 Features

- Meets automotive application requirements
- Simple solution for sequencing 3 power rails with a single input signal
- Easily cascade up to 3 devices to sequence up to 9 power rails
- Supports both power-up and power-down control
- Compact 2.9mm x 1.9mm package
- Low 25 μ A quiescent current
- Input voltage range: 2.7V to 5.5V
- Standard timing options available

2 Applications

- Advanced Driver Assistance Systems (ADAS)
- Automotive Camera Modules
- Security Cameras
- Servers
- Networking Products
- FPGA Power Sequencing
- Microprocessor and Microcontroller Power Sequencing
- Multi-Power Sequencing

3Description

LM3880This simple power sequencing controller offers an easy way to manage the power-up and power-down order for multiple independent voltage rails. By staggering the start-up sequence, it helps prevent latch-up issues or large inrush currents that could affect system reliability.

This straightforward sequencer comes in a 6-pin SOT-23 package, featuring a precision enable pin and three open-drain status outputs. These outputs can be pulled up to power supplies that are different from the sequencer's V_{DD} (as long as they do not exceed the recommended max voltage, which is 0.3V above V_{DD}), making it compatible with ICs requiring different enable signal levels. When LM3880 is enabled, the three status outputs are released one after another at defined delays, allowing connected power supplies to start up in order. During power-down, the outputs follow the reverse sequence to help prevent latch-up.

The EPROM feature lets you fully customize each delay and sequence. For non-standard setups, please contact Texas Instruments (TI).

Device Information(1)

Device Model	Package	Package Dimensions
LM3880	DBV SOT (6)	2.90mm × 1.60mm (Nominal)

(1)For all available packages, please see the ordering appendix at the end of the datasheet.

Simple Power Sequencing Control

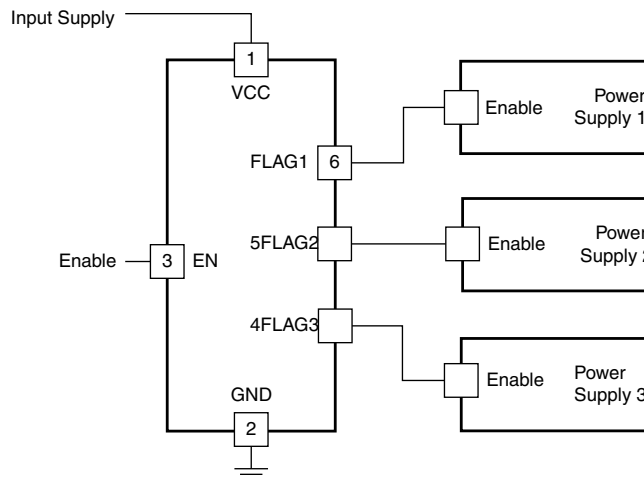


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4 Revision History

Note: Page numbers in previous versions may differ from those in this edition.

Updates from Revision K (February 2016) to Revision L

Page

• Revised the "Features" section to specify how many voltage rails can be sequenced using a single device	1
• Added support for cascading multiple devices	1
• Device dimensions now specified in the "Features" section.....	1
• The "Applications" section now includes FPGA power sequencing control.....	1
• Added information about the open-drain FLAG pin in the "Description" section	1
• Included an "I/O" column in the Pin Functions table	3
• Renamed the Part Numbering Rules section to Device Naming Rules	18

Updates from Revision J (December 2014) to Revision K

Page

• Changed the handling ratings section to ESD ratings and moved storage temperature under absolute maximum ratings	4
• Removed the "Custom Sequencing and Order" section	12
• Added cross-references to the timing diagrams	18

Updates from Revision I (March 2013) to Revision J

Page

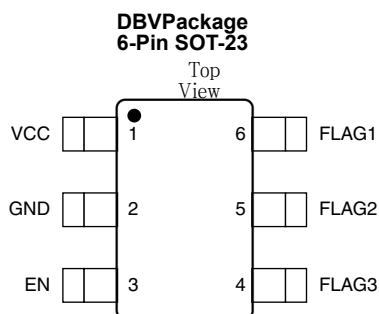
• Added tables for handling ratings, feature descriptions, device functional modes, applications and implementation, power supply recommendations, layout guidance, device and documentation support, and mechanical, packaging, and ordering information.....	4
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Updates from Revision H (March 2013) to Revision I

Page

• Reformatted the datasheet layout from National Semiconductor format to TI format.	18
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5 Pin Configuration and Functions



Pin Functions

Pin		I/O(1)	Description
Name	Number		
EN	3	I	Precision Enable Pin
FLAG1	6	O	Open-Drain Output 1
FLAG2	5	O	Open-Drain Output 2
FLAG3	4	O	Open-Drain Output 3
Ground	2	G	Ground
VCC	1	I	Power Input

(1) I =Input, O =Output, G =Ground

LM3880

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6 Specifications

6.1 Absolute Maximum Ratings

Measured under natural airflow unless otherwise specified⁽¹⁾⁽²⁾

	Minimum	Maximum	Unit
VCC	-0.3	6	V
EN,FLAG1,FLAG2,FLAG3	-0.3	6	V
FLAGMaximum Output Current		50	mA
Maximum Junction Temperature		150	°C
Lead Temperature (during soldering,5s)		260	°C
Storage Temperature T _{stg}	-65	150	°C

(1) Stresses that exceed the absolute maximum ratings listed may result in permanent damage to the device. These ratings are intended only for operation under specified conditions, and functionality or operation beyond these recommended conditions is not guaranteed. Prolonged exposure to absolute maximum ratings may affect device reliability.

(2) For military or aerospace-specific devices, please contact Texas Instruments (TI) sales offices or authorized distributors for availability and technical specifications.

6.2 ESD Ratings

	Value	Unit
V(ESD) Electrostatic Discharge Human Body Model (HBM), compliant with ANSI/ESDA/JEDEC JS-001, all pins ⁽¹⁾	±2	kV

(1) JEDEC Document JEP155 states: At 500V HBM, safe manufacturing is possible under standard ESD control procedures.

6.3 Recommended Operating Conditions

Measured in ambient air temperature range (unless otherwise noted)

	Minimum	Maximum	Unit
VCC to GND	2.7	5.5	V
EN,FLAG1,FLAG2,FLAG3	-0.3	V _{CC} +0.3	V
Junction Temperature	-40	125	°C

6.4 Thermal Characteristics

Thermal Metrics ⁽¹⁾	LM3880	Unit
	DBV (SOT-23)	
	6 pins	
R _{θJA} Junction-to-ambient thermal resistance	187.6	°C/W
R _{θJC(top)} Junction-to-case (top) thermal resistance	127.4	°C/W
R _{θJB} Junction-to-board thermal resistance	31.5	°C/W
ψ _{JT} Junction-to-top characterization parameter	23.3	°C/W
ψ _{JB} Junction-to-board characterization parameter	31.0	°C/W

(1) For more details on traditional and new thermal metrics, see IC Package Thermal Metrics Application Report([Document No.: SPRA953](#)).

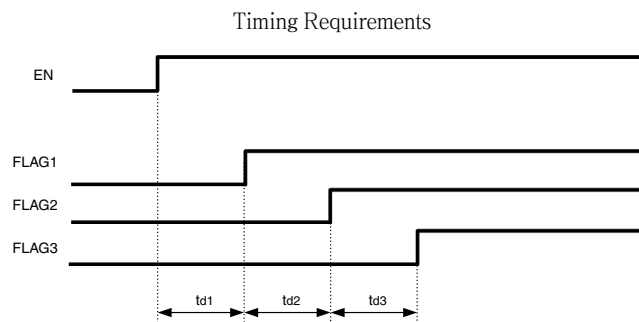
6.5 Electrical Characteristics

Unless otherwise noted, limits apply to all timing options and $V_{CC} = 3.3V$. Minimum and maximum values are specified over the full operating temperature range ($T_J = -40^\circ\text{C}$ to $+125^\circ\text{C}$), and are defined by test, design, or statistical correlation. Typical values indicate the most probable parameter value at $T_J = 25^\circ\text{C}$ and are for reference only.

Parameter	Test Conditions	Min ⁽¹⁾	Typ ⁽²⁾	Max ⁽¹⁾	Unit
I_{Q} Open-Drain FLAG Indicator	Quiescent Operating Current		25	80	μA
I_{FLAG}	FLAGx Leakage Current	$V_{FLAGx} = 3.3V$		1	20
					nA
V_{OL} Power-Up Sequence	FLAGx Low-Level Output	$I_{FLAGx} = 1.2\text{mA}$		0.4	V
t_{d1}	Timer Delay 1 Accuracy	All Other Timing Options	-15%	15%	
		2ms Timing Option	-20%	20%	
t_{d2}	Timer Delay 2 Accuracy	All Other Timing Options	-15%	15%	
		2ms Timing Option	-20%	20%	
t_{d3}	Timer Delay 3 Accuracy	All Other Timing Options	-15%	15%	
		2ms Timing Option	-20%	20%	
Power-Down Sequence					
t_{d4}	Timer Delay 4 Accuracy	All Other Timing Options	-15%	15%	
		2ms Timing Option	-20%	20%	
t_{d5}	Timer Delay 5 Accuracy	All Other Timing Options	-15%	15%	
		2ms Timing Option	-20%	20%	
t_{d6}	Timer Delay 6 Accuracy	All Other Timing Options	-15%	15%	
		2ms Timing Option	-20%	20%	
Timing Delay Error					
$(t_{d(x)} - 400\mu\text{s})/t_{d(x+1)}$	Timing Delay Ratio	For x = 1 or 4	95%	105%	
		For x = 1 or 4, 2ms Option	90%	110%	
$t_{d(x)}/t_{d(x+1)}$	Timing Delay Ratio	For x = 2 or 5	95%	105%	
		For x = 2 or 5, 2ms Option	90%	110%	
Enable Pin					
V_{EN}	EN Pin Threshold		1.0	1.25	1.4
					V
I_{EN}	EN Pin Pull-up Current	$V_{EN} = 0V$	7		μA

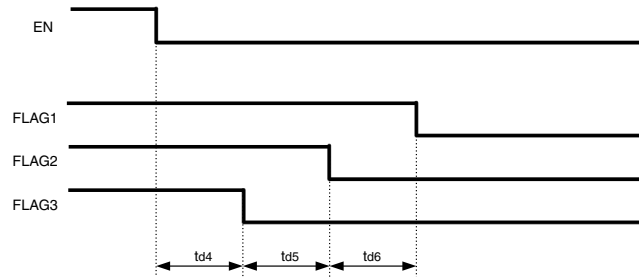
(1) All limits are 100% production tested at 25°C . Statistical quality control (SQC) methods are used to ensure limits across the full operating temperature range by correlation. These limits are used to calculate TI's Average Outgoing Quality Level (AOQL).

(2) Typical values are measured at 25°C and represent the most probable standard for the parameter.



All standard options use sequence 1 to indicate the output flag's rise and fall order. For more details on alternate sequences, see Section 11.1.2.

Figure 1. Power-Up Sequence



All standard options use sequence 1 to indicate the order in which output flags rise and fall. For more details on possible sequence variations, please refer to Section 11.1.2.

Figure 2. Power-down sequencing diagram

6.6 Typical Characteristics

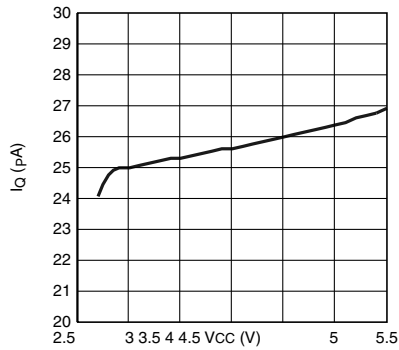


Figure 3. Quiescent Current vs VCC Relationship

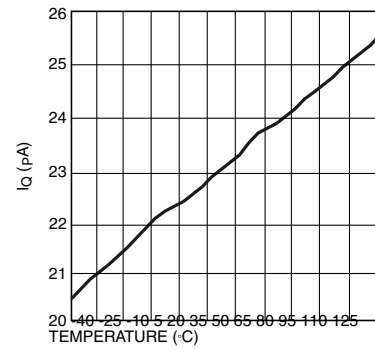


Figure 4. Quiescent Current vs Temperature (VCC = 3.3V)

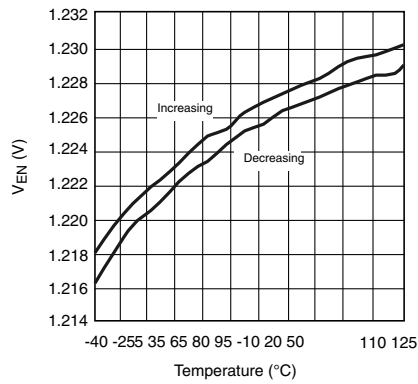


Figure 5. Enable Threshold vs Temperature

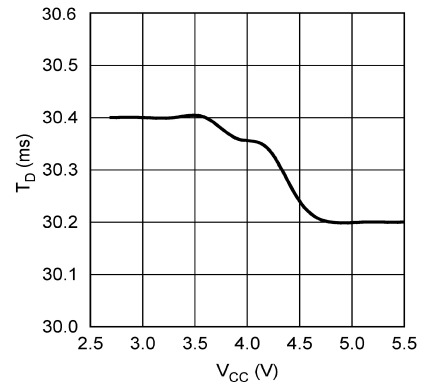


Figure 6. Delay Time (30ms) vs VCC Relationship

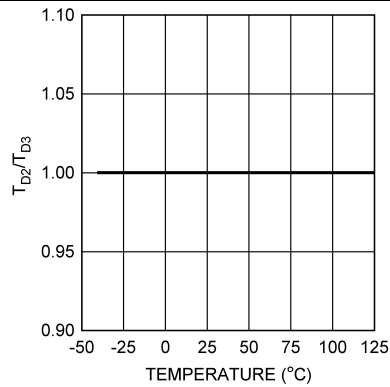


Figure 7. Delay Rate vs Temperature

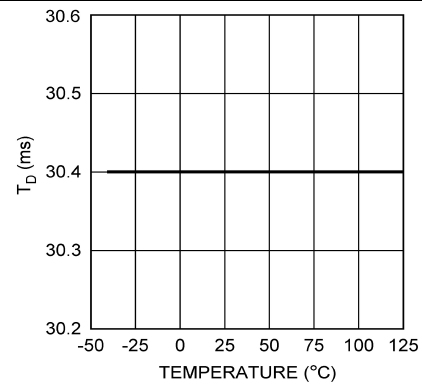


Figure 8. Delay Time (30ms) vs Temperature

Typical Characteristics (continued on next page)

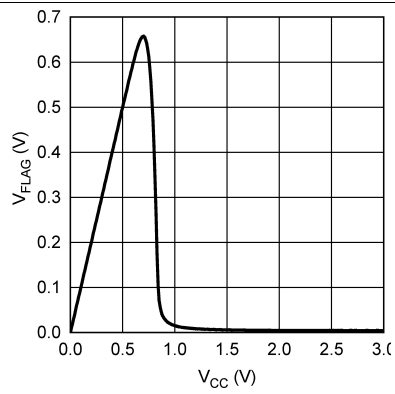


Figure 9. Relationship between V_{OL} and V_{CC} ($R_{FLAG} = 100k\Omega$)

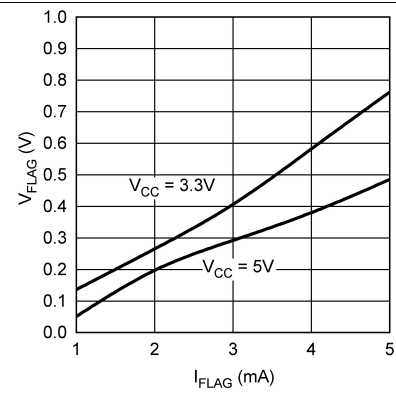


Figure 10. Relationship Between Output Voltage and Output Current

7 Detailed Description

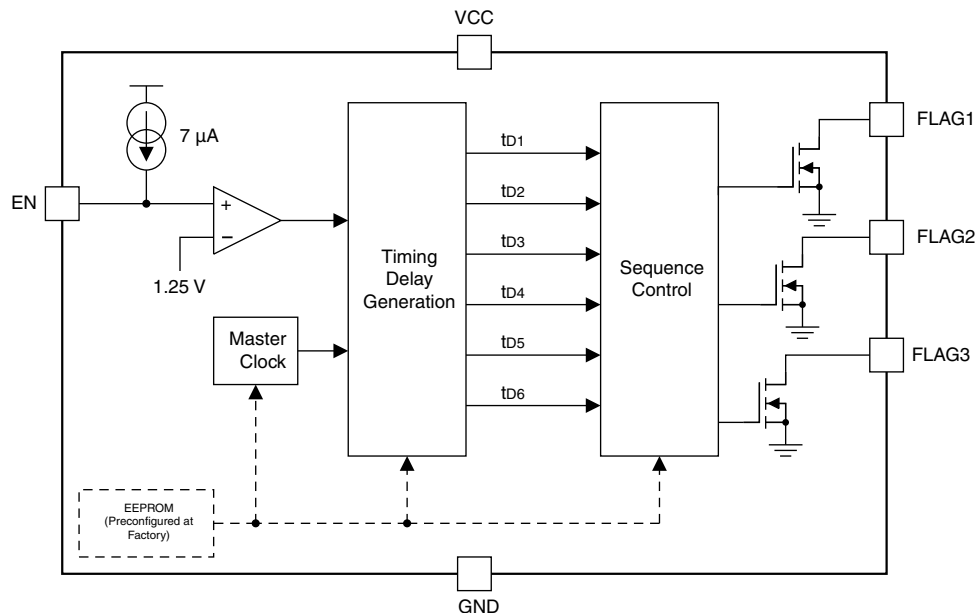
7.1 Overview

LM3880 is a straightforward power sequencing controller designed to easily manage the timing of multiple power rails. It features six independent timers to control the sequencing of three open-drain output flags (for both power-up and power-down). These flags can be connected to the shutdown or enable pins of linear regulators and switching power supplies, providing precise control over power operations. This system makes it possible to design a complete power solution without worrying about potential issues like large inrush currents or latch-up events.

The device's sequencing is fully managed by the enable (EN) pin. After power-up, all flags remain low until the EN pin is pulled high. Once EN is set, the power-up sequence begins. Before the initial fixed delay ends, the first flag (FLAG1) stays low; after it is released, another timer controls the delay before the second flag (FLAG2) is released. This process repeats until all three flags have been released in order.

The power-down sequence mirrors the power-up process, but in reverse. When the EN pin is de-asserted, a timer is triggered to delay the deactivation of the third flag (FLAG3). After that, the second and first flags follow in sequence, each with their own delays. The three timers for the power-down sequence can be programmed independently and operate separately from the power-up timers.

7.2 Functional Block Diagram



7.3 Feature Description

7.3.1 Enable Pin Operation

The timing of the LM3880 is controlled by setting the enable signal. The enable pin features an internal comparator referenced to a 1.25V bandgap voltage, ensuring precise threshold detection. This allows you to use an external capacitor for delay control, or to start sequencing based on specific events (such as line voltage reaching 90% of its nominal value). To add an extra delay to the VCC rail, simply connect a capacitor to the EN pin as shown in [Figure 11](#).

Feature Description (Continued on next page)

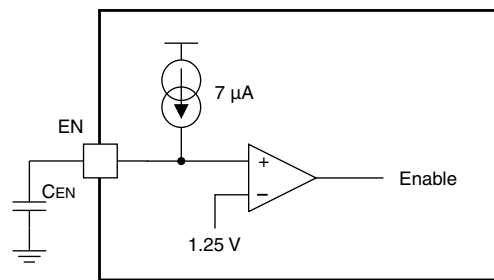


Figure 11. Capacitor Timing Sequence

When using the internal pull-up current source to charge the external capacitor (C_{EN}), you can use [Equation 1](#) to calculate the enable pin delay:

$$t_{enable_delay} = \frac{1.25V \times C_{EN}}{7 \mu A} \quad (1)$$

A resistor divider can also be used to enable the device based on a specific voltage threshold. Be sure to consider the effect of the internal current source when selecting resistor values.

One of the features of the EN pin is its ability to operate without interference. The first timer starts counting at the rising threshold, but it always resets if the EN pin is cleared before the first output flag is released. [Figure 12](#) illustrates this:

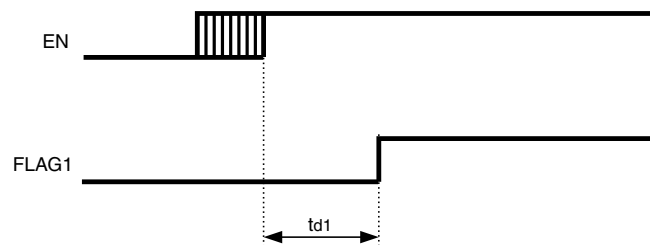


Figure 12. EN Interference

7.3.2 Incomplete Sequence Operation

If the enable signal stays high for the entire power-up sequence, the device will operate as shown in the standard timing diagram. However, if the enable signal is cleared before the sequence finishes, the device enters a controlled shutdown state. This allows the system to perform controlled power cycling and prevents any latch-up. This state only occurs if the enable pin is cleared after timer 1 completes but before the entire power-up sequence finishes.

When this event occurs, the falling edge on the EN pin will reset the current timer and allow the remaining power-up cycles to finish before starting the power-down sequence. The power-down sequence begins about 120ms after the last power-up flag. This keeps all voltage rail outputs stable until system power-down. [Figure 13](#) shows an example of this process:

Product Features (Continued on Next Page)

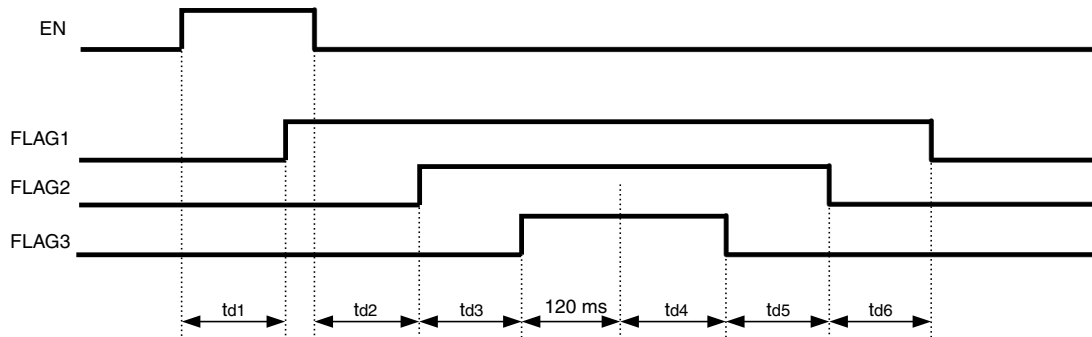


Figure 13. Incomplete Power-Up Sequence and Complete Power-Up Sequence

When the enable signal is de-asserted, the device initiates its power-down sequence. If the enable signal goes high before the power-down sequence is finished, the device guarantees the power-down process completes before starting to power up. This ensures the system doesn't partially power down or up, helping prevent latch-up events (such as in FPGAs and microprocessors). This condition only occurs if the enable pin is brought high after Timer 1 finishes, but before the entire power-down sequence completes.

When this occurs, the rising edge on the enable pin resets the current timer, allowing the remaining power-down period to finish before the power-up sequence begins. The power-up sequence starts about 120ms after the last power-down flag. This lets all voltage rails discharge before the system powers up. Figure 14 shows an example of this operation:

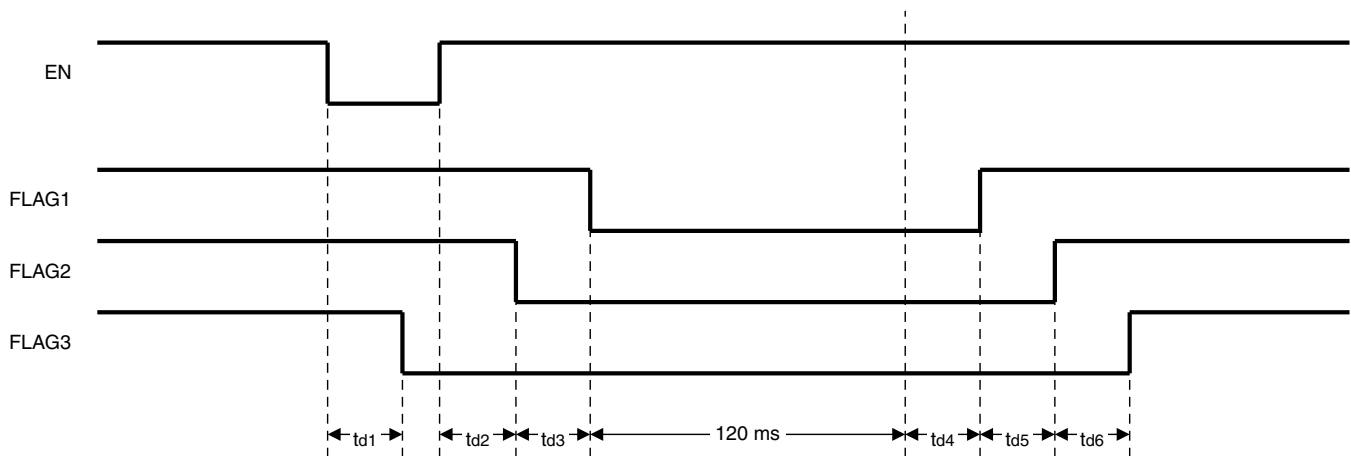


Fig. 14. Incomplete Power Down Sequence & Complete Power Down Sequence

All internal timers are generated by a master clock with an ultra-low temperature coefficient. This delivers high accuracy across the full temperature range and maintains consistent ratios between timers. Timers 1 and 4 have an additional delay of about 400μs due to EPROM refresh. This refresh time has a negligible effect compared to all timer delays, except for the shortest timer hold.

7.4 Device Functional Modes

7.4.1 Power-Up via EN Pin

The power-up sequence of this simple power supply sequencer is fully managed by the enable (EN) pin. After power is applied, all flags remain low until the precision enable pin is pulled high. Once the EN pin is set, the power-up sequence begins.

7.4.2 Powering down via EN pin control

When the EN pin is cleared, the power-down sequence starts. A timer is triggered, which delays the deactivation of the third flag (FLAG3). Afterward, the second and first flags follow in order, each after its respective delay.

8 Applications and Implementation

Note: The information in the following applications section is outside the scope of TI device specifications, and TI does not guarantee its accuracy or completeness. TI Customers are responsible for determining whether the device fits their needs. Designs must be validated and tested to ensure proper system operation.

8.1 Application Information

8.1.1 Open-Drain Status Pull-Up

The simple power sequence controller has three open-drain status outputs that require pull-up resistors to operate properly. A 100k pull-up resistor can be used.

8.1.2 Device Enable

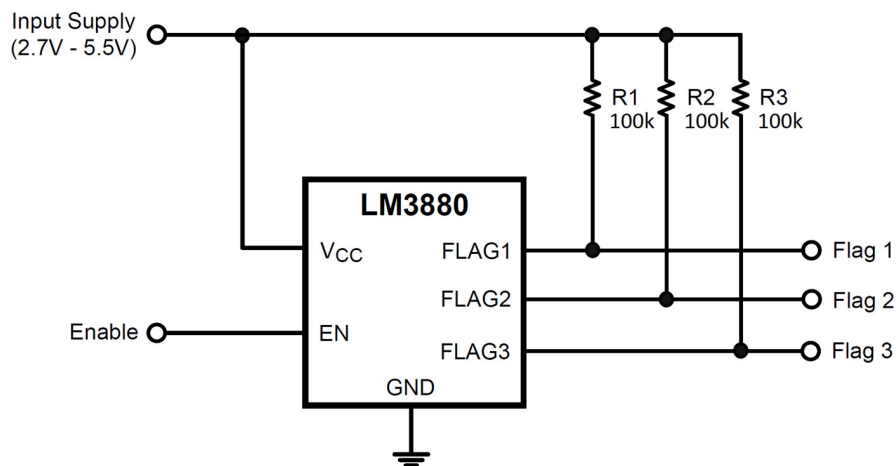
See [Enable Pin Operation](#)

8.2 Typical Applications

8.2.1 Simple Power Sequence for Three Power Rails

The basic power sequencer controls the power-up and power-down order for three power rails.

For example, the LM3880 sequence 1 (orderable part number LM3880MF-1AA) features a power-up sequence (1 - 2 - 3) and a power-down sequence (3 - 2 - 1). For other sequence options, see [Table 3](#) and [Table 4](#), or contact TI for additional options.



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Figure 15. Typical Application Circuit

Typical Application (continued)

8.2.1.1 Design Requirements

This design example uses the parameters listed in [Table 1](#) as input values. The circuit in [Figure 15](#) can support various power-down sequences depending on device programming. For more information on power-down sequence options, see [Table 3](#).

Table 1. Design Parameters

Design Parameter	Example Value
Input Supply Voltage Range	2.7V to 5.5V
Status Output Voltage, ENHigh Level	Input Power Supply
Status Output Voltage, ENLow Level	0V
Status Timing Delay	30ms
Power-Up Sequence	1-2-3
Power-Down Sequence	3-2-1

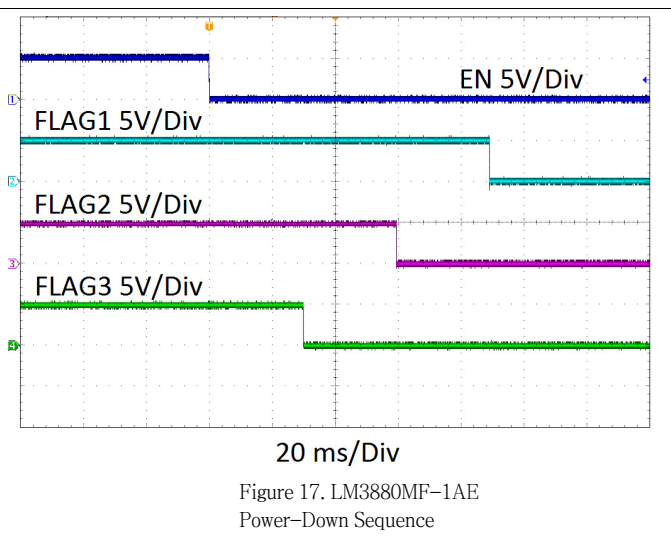
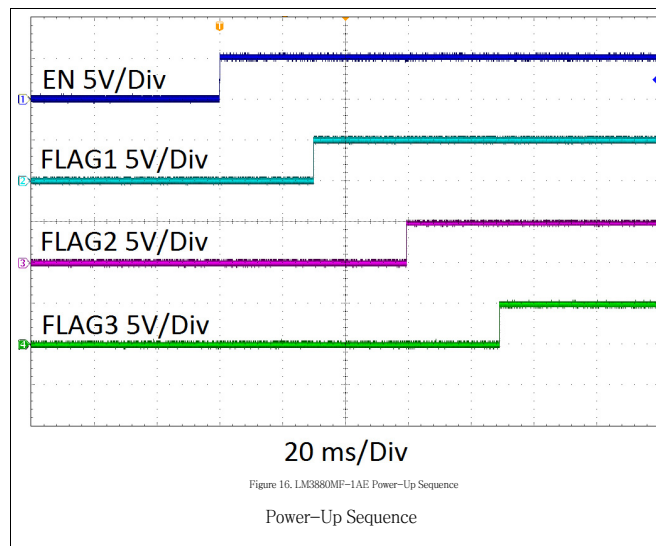
8.2.1.2 Detailed Design Flow

Table 2. Bill of Materials

Reference	Description	Component	Quantity	Manufacturer
U1	LM3880, Sequence 1, 30ms timing	LM3880	1	Texas Instruments (TI)
R1	100kΩ resistor, 0603	CRCW0603100KFKEA	1	Vishay
R2	100kΩ resistor, 0603	CRCW0603100KFKEA	1	Vishay
R3	100kΩ resistor, 0603	CRCW0603100KFKEA	1	Vishay

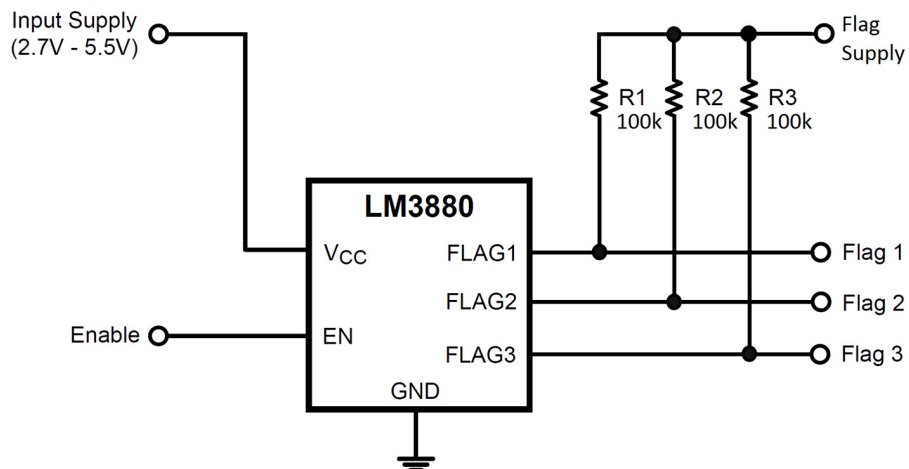
This application uses a basic power sequencing controller with Sequence 1 and a 30ms timing option. For more details about the sequence and timing options, refer to [Application Waveform](#).

8.2.1.3 Application Waveform



8.2.2 Sequencing Control Using Separate Indicator Voltage Levels

For applications where the indicator output voltage differs from VCC, a separate indicator power supply can be used to pull up the open-drain output of a basic power sequencing controller. This is especially helpful when the indicator output needs to connect to an input voltage that isn't the same as VCC. Designers must ensure that the indicator power supply voltage does not exceed [the recommended operating conditions](#) of VCC + 0.3V.



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Figure 18. Timing control using a separate flag power supply

8.3 Important Notes

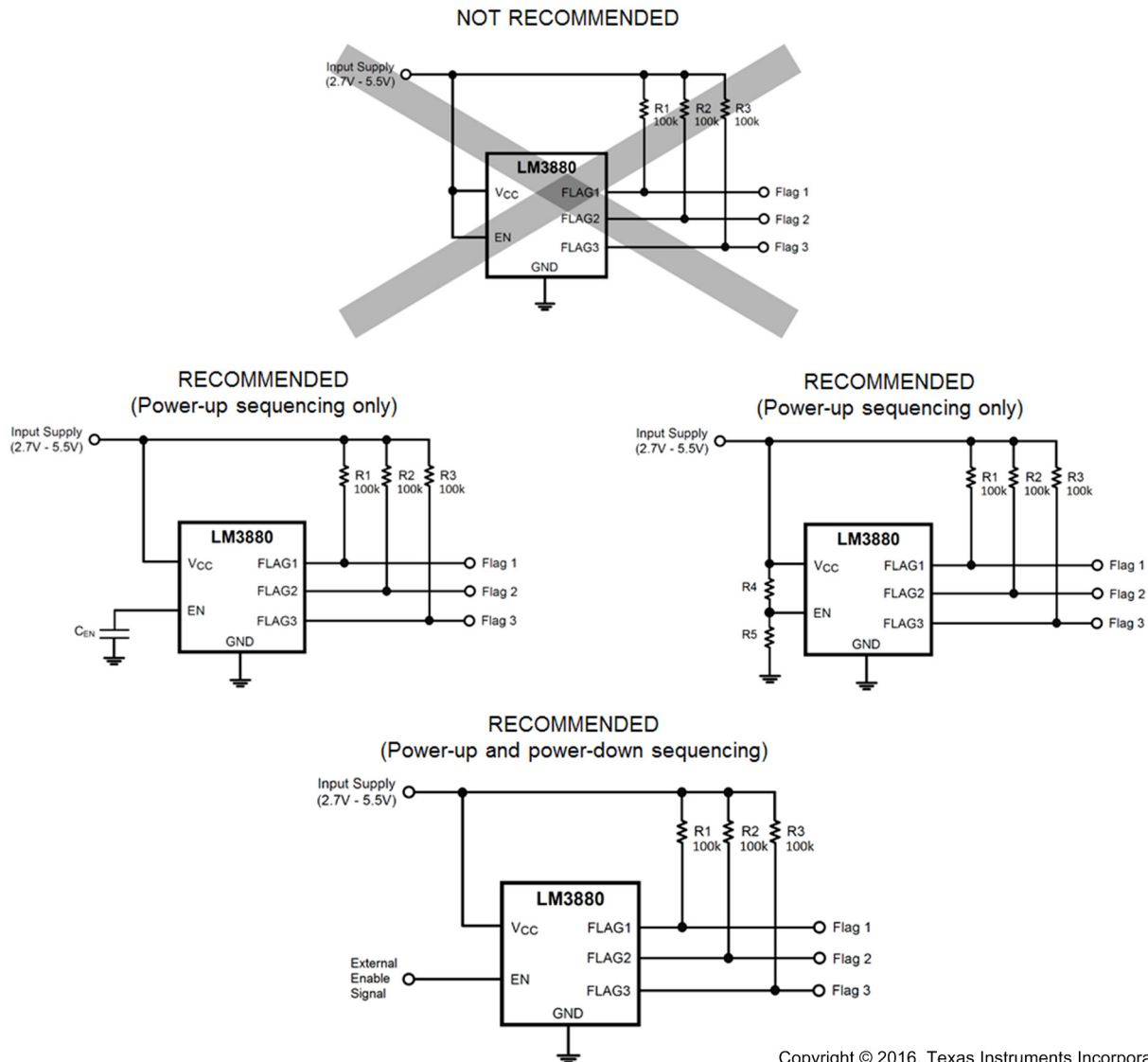
It is recommended not to connect the EN pin directly to VCC. During power-up, EN voltage should stay below the EN threshold until VCC rises above the minimum operating voltage. If you connect EN to VCC, this guideline is broken and the flag output may behave unpredictably, especially when VCC is ramping up slowly. For systems that only need power-up sequencing, you can add a capacitor to the EN pin for delay, or use a resistor divider to trigger the device based on a specific voltage threshold. While these solutions work for powering up, they won't provide controlled power-down sequencing—the flag output will simply track the input supply. For applications requiring both power-up and power-down sequencing, use an external enable signal (such as a GPIO from a microcontroller) to properly manage the flag output status.

LM3880

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Note (continued on next page)



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Figure 19. Recommended EN Connection

9 Power Supply Recommendations

VCC pins should be placed as close as possible to the power input (2.7–5.5V). An input capacitor isn't required, but if there's noise at the VCC pin, it's recommended to add one. A 0.1µF ceramic capacitor can be used to filter out any unwanted noise.

10 Board Layout

10.1 Layout Guidelines

- Install a pull-up resistor between the status output pin and the main power input (typically VCC). You may also use a separate status power supply. For best performance, keep these resistors as close as possible to the power sequencer and status supply. Use the shortest possible traces for connections. A typical pull-up resistor value is 100kΩ.
- For applications with tight timing requirements, use short and equal-length wires to connect the status output to its input. This helps minimize propagation delay and timing errors between status outputs along the trace.

10.2 Layout Examples

Figures 20 and 21 show example layouts for the LM3880. These examples are taken from the LM3880EVAL.

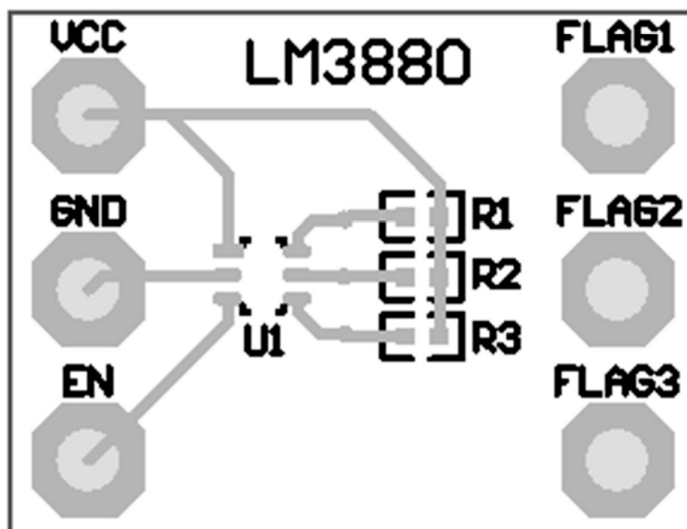


Figure 20. LM3880 Top View

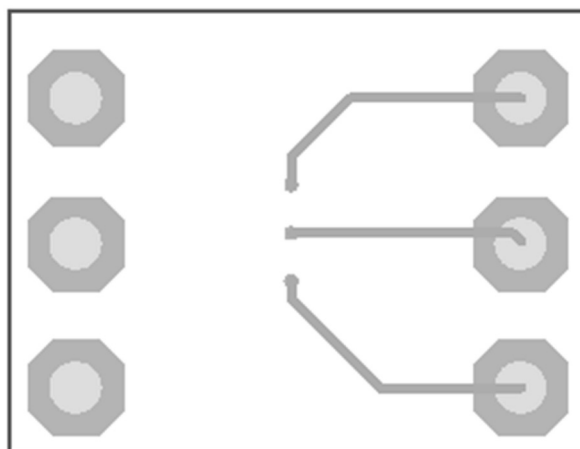


Figure 21. LM3880 Bottom View

11 Device and Documentation Support

11.1 Device Support

11.1.1 Third-Party Products Disclaimer

Information published by TI relating to third-party products or services does not constitute endorsement or warranty of the suitability of such products or services, whether alone or in combination with any TI products or services.

11.1.2 Device Naming Conventions

A list of orderable parts is provided in the Packaging Options Appendix.

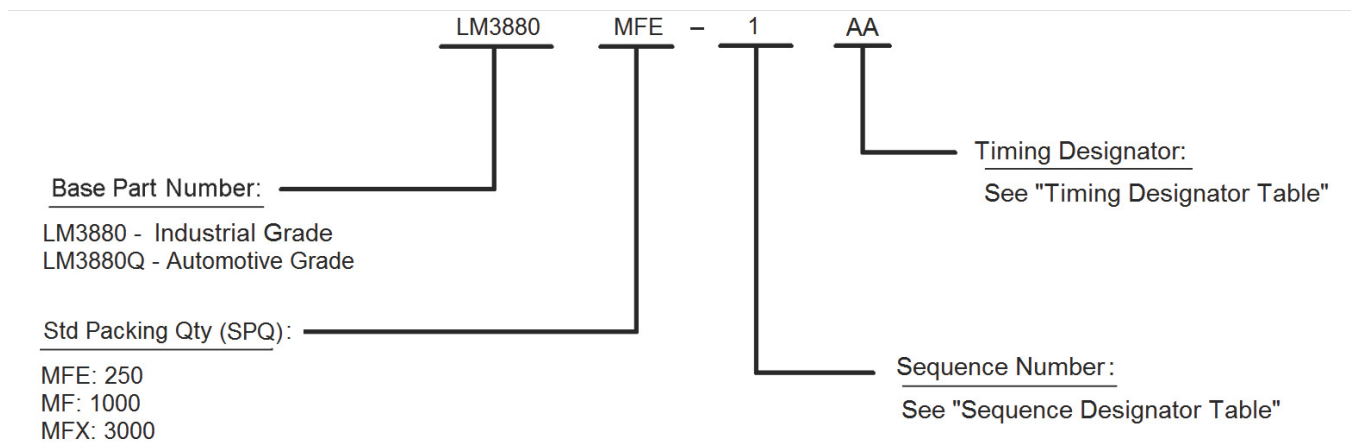


Figure 22. Device Naming Conventions

Table 3. Sequence Indicator Table (1)

Sequence Number	Flag Order	
	Power Up	Power Down
1	1-2-3	3-2-1
2	1-2-3	3-1-2
3	1-2-3	2-3-1
4	1-2-3	2-1-3
5	1-2-3	1-3-2
6	1-2-3	1-2-3

(1) See Figure 1 and Figure 2.

Table 4. Timing Indicator Table(1)

Timing Indicator	Delay (ms)					
	td1	td2	td3	td4	td5	td6
AA	10	10	10	10	10	10
AB	30	30	30	30	30	30
AC	60	60	60	60	60	60
AD	120	120	120	120	120	120
AE	2	2	2	2	2	2
AF	16	16	16	16	16	16

(1) See Figure 1 and Figure 2.

11.2 Community Resources

The following links connect you to TI community resources. Content provided by distributors is offered “as is.” These materials do not represent TI technical specifications and may not reflect TI’s position; please refer to TI’s [Terms of Use](#).

[TI E2E™Online Community](#) TI engineers supporting engineers (E2E) Community. This space is designed to foster collaboration among engineers. At e2e.ti.com, you can ask questions, share your expertise, gain new insights, and work with fellow engineers to solve challenges.

[Design Support](#) TI Reference design support from TI helps you quickly find helpful E2E forums, design support tools, and technical support contact details.

11.3 Trademarks

E2E is a trademark of Texas Instruments.

11.4 Electrostatic Discharge (ESD) Warning



This device is not ESD protected. When storing or handling, be sure to clip the leads together or place the device on conductive foam to prevent MOS gates from being damaged by ESD electricity.

11.5 Glossary of Terms

[SLYZ022](#) — TI Glossary.

This glossary lists and explains terms, abbreviations, and definitions.

12 Mechanical, Packaging, and Ordering Information

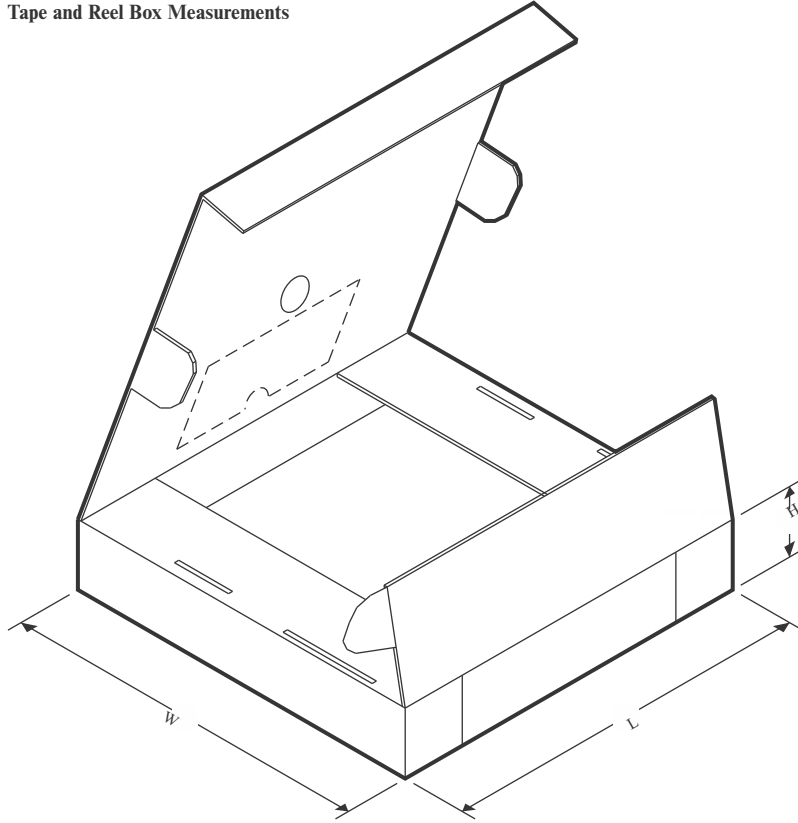
The following pages provide mechanical, package, and ordering details. This is the most current information available for the specified device. Information is subject to change without notice, and this document will not be updated. For the browser version of this datasheet, refer to the navigation bar on the left.

Packaging Details

Orderable Part Number	Status (1)	Package Style	Package Outline	Number of Pins	Quantity per Package	Environmental Plan (2)	Lead Finish / Ball Material (6)	MSL Peak Temperature (3)	Operating Temperature (°C)	Device Marking (4/5)	Samples Available
LM3880MF-1AA	NRND	SOT-23	DBV	6 pins	Pack of 1000	Non-RoHS, Green compliant	Contact TI for details	Level 1, 260°C maximum, unlimited shelf life	Operating temperature: -40°C to 125°C	Marking: F20A	
LM3880MF-1AA/NOPB part	Currently available	SOT-23	DBV	6 pins	Pack of 1000	RoHS certified, Green compliant	Mark: SN	Level 1, 260°C maximum, unlimited shelf life	Operating temperature: -40°C to 125°C	Marking: F20A	Samples
LM3880MF-1AB/NOPB part	Currently available	SOT-23	DBV	6 pins	Pack of 1000	RoHS certified, Green compliant	Mark: SN	Level 1, 260°C maximum, unlimited shelf life	Operating temperature: -40°C to 125°C	Marking: F21A	Samples
LM3880MF-1AC/NOPB part	Currently available	SOT-23	DBV	6 pins	Pack of 1000	RoHS certified, Green compliant	Mark: SN	Level 1, 260°C maximum, unlimited shelf life	Operating temperature: -40°C to 125°C	Marking: F22A	Samples
LM3880MF-1AD/NOPB part	Currently available	SOT-23	DBV	6 pins	Pack of 1000	RoHS certified, Green compliant	Mark: SN	Level 1, 260°C maximum, unlimited shelf life	Operating temperature: -40°C to 125°C	Marking: F23A	Samples
LM3880MF-1AE/NOPB part	Currently available	SOT-23	DBV	6 pins	Pack of 1000	RoHS certified, Green compliant	Mark: SN	Level 1, 260°C maximum, unlimited shelf life	Operating temperature: -40°C to 125°C	F25A	Samples
LM3880MF-1AF/NOPB	Currently available	SOT-23	DBV	6 pins	Pack of 1000	RoHS certified, Green compliant	Mark: SN	Level 1, 260°C maximum, unlimited shelf life	Operating temperature: -40°C to 125°C	F31A	Samples
LM3880MFE-1AA/NOPB	Currently available	SOT-23	DBV	6 pins	250	RoHS certified, Green compliant	Mark: SN	Level 1, 260°C maximum, unlimited shelf life	Operating temperature: -40°C to 125°C	Marking: F20A	Samples
LM3880MFE-1AB/NOPB	Currently available	SOT-23	DBV	6 pins	250	RoHS certified, Green compliant	Mark: SN	Level 1, 260°C maximum, unlimited shelf life	Operating temperature: -40°C to 125°C	Marking: F21A	Samples
LM3880MFE-1AC/NOPB	Currently available	SOT-23	DBV	6 pins	250	RoHS certified, Green compliant	Mark: SN	Level 1, 260°C maximum, unlimited shelf life	Operating temperature: -40°C to 125°C	Marking: F22A	Samples
LM3880MFE-1AD/NOPB	Currently available	SOT-23	DBV	6 pins	250	RoHS certified, Green compliant	Mark: SN	Level 1, 260°C maximum, unlimited shelf life	Operating temperature: -40°C to 125°C	Marking: F23A	Samples
LM3880MFE-1AE/NOPB	Currently available	SOT-23	DBV	6 pins	250	RoHS certified, Green compliant	Mark: SN	Level 1, 260°C maximum, unlimited shelf life	Operating temperature: -40°C to 125°C	F25A	Samples
LM3880MFE-1AF/NOPB	Currently available	SOT-23	DBV	6 pins	250	RoHS certified, Green compliant	Mark: SN	Level 1, 260°C maximum, unlimited shelf life	Operating temperature: -40°C to 125°C	F31A	Samples
LM3880MFX-1AA/NOPB	Currently available	SOT-23	DBV	6 pins	3000	RoHS certified, Green compliant	Mark: SN	Level 1, 260°C maximum, unlimited shelf life	Operating temperature: -40°C to 125°C	Marking: F20A	Samples
LM3880MFX-1AB/NOPB	Currently available	SOT-23	DBV	6 pins	3000	RoHS certified, Green compliant	Mark: SN	Level 1, 260°C maximum, unlimited shelf life	Operating temperature: -40°C to 125°C	Marking: F21A	Samples
LM3880MFX-1AC/NOPB	Currently available	SOT-23	DBV	6 pins	3000	RoHS certified, Green compliant	Mark: SN	Level 1, 260°C maximum, unlimited shelf life	Operating temperature: -40°C to 125°C	Marking: F22A	Samples
LM3880MFX-1AD/NOPB	Currently available	SOT-23	DBV	6 pins	3000	RoHS certified, Green compliant	Mark: SN	Level 1, 260°C maximum, unlimited shelf life	Operating temperature: -40°C to 125°C	Marking: F23A	Samples
LM3880MFX-1AE/NOPB	Currently available	SOT-23	DBV	6 pins	3000	RoHS certified, Green compliant	Mark: SN	Level 1, 260°C maximum, unlimited shelf life	Operating temperature: -40°C to 125°C	F25A	Samples
LM3880MFX-1AF/NOPB	Currently available	SOT-23	DBV	6 pins	3000	RoHS certified, Green compliant	Mark: SN	Level 1, 260°C maximum, unlimited shelf life	Operating temperature: -40°C to 125°C	F31A	Samples

(1)The definitions for marketing status values are as follows:

Tape and Reel Box Measurements



*All measurements are approximate

Product	Package Format	Package Diagram	Number of Pins	Units per Package	Length (mm)	Width (mm)	Height (mm)
LM3880MF-1AA	SOT-23	DBV	6	1000	208.0	191.0	35.0
LM3880MF-1AA/NOPB	SOT-23	DBV	6	1000	208.0	191.0	35.0
LM3880MF-1AB/NOPB	SOT-23	DBV	6	1000	208.0	191.0	35.0
LM3880MF-1AC/NOPB	SOT-23	DBV	6	1000	208.0	191.0	35.0
LM3880MF-1AD/NOPB	SOT-23	DBV	6	1000	208.0	191.0	35.0
LM3880MF-1AE/NOPB	SOT-23	DBV	6	1000	208.0	191.0	35.0
LM3880MF-1AF/NOPB	SOT-23	DBV	6	1000	208.0	191.0	35.0
LM3880MFE-1AA/NOPB	SOT-23	DBV	6	250	208.0	191.0	35.0
LM3880MFE-1AB/NOPB	SOT-23	DBV	6	250	208.0	191.0	35.0
LM3880MFE-1AC/NOPB	SOT-23	DBV	6	250	208.0	191.0	35.0
LM3880MFE-1AD/NOPB	SOT-23	DBV	6	250	208.0	191.0	35.0
LM3880MFE-1AE/NOPB	SOT-23	DBV	6	250	208.0	191.0	35.0
LM3880MFE-1AF/NOPB	SOT-23	DBV	6	250	208.0	191.0	35.0
LM3880MFX-1AA/NOPB	SOT-23	DBV	6	3000	208.0	191.0	35.0
LM3880MFX-1AB/NOPB	SOT-23	DBV	6	3000	208.0	191.0	35.0
LM3880MFX-1AC/NOPB	SOT-23	DBV	6	3000	208.0	191.0	35.0
LM3880MFX-1AD/NOPB	SOT-23	DBV	6	3000	208.0	191.0	35.0
LM3880MFX-1AE/NOPB	SOT-23	DBV	6	3000	208.0	191.0	35.0

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